Austing (Lianghan) Dong

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Research Interests

Machine learning and multimodal models, with a focus on understanding and steering model behavior: interpretability of vision–language model reasoning (attention-guided visualization), empirical analysis of model specialization—how the gap between few-shot prompting and fine-tuning scales with model size and pretraining—and interactive, human-in-the-loop training of VLMs.

Education

University of Waterloo Sep 2022 – present (expected 2027)
Bachelor of Computer Science (Co-op) Waterloo, ON, Canada

- GPA: 3.7/4.0.
- President's Scholarship of Distinction; Faculty of Math Entrance Scholarship ($25,000 CAD).

Publications

1. **Lianghan Dong**, Anamaria Crisan. "Probing the Visualization Literacy of Vision Language Models: the Good, the Bad, and the Ugly." *IEEE Transactions on Visualization and Computer Graphics (TVCG), Special Issue for IEEE VIS 2025*. [paper] [code]
2. **Lianghan Dong**. "Enhancing Image Segmentation for ICH through Transfer Learning from Stroke MRI to ICH CT." *Proceedings of ICIAAI 2024*. [paper]

Research Experience

Undergraduate Research Fellow, University of Waterloo May 2026 – Present
Advisor: Prof. Yuntian Deng Waterloo, Canada

- Investigating the performance gap between few-shot prompting and fine-tuning for specializing language models, and whether scaling model size or improving pretraining closes it.
- Benchmarking open model families (GPT-2, Qwen) across sizes on a multi-task benchmark to chart the gap as a function of scale and pretraining recency.

Undergraduate Research Fellow, University of Waterloo Sep 2025 – Dec 2025
Advisor: Prof. Yuntian Deng Waterloo, Canada

- Explored fine-tuning vision–language models with attention regularization and human-labeled supervision to improve controllability and visual interpretability.
- Implemented attention visualization and labeling interfaces for multimodal transformer layers, analyzing head-wise token alignment across image–text pairs.

Undergraduate Research Fellow, University of Waterloo Jan 2025 – Apr 2025
Advisor: Prof. Anamaria Crisan Waterloo, Canada

- First author of a paper accepted to IEEE TVCG (Special Issue for IEEE VIS 2025), introducing AG-CAM, an attention-guided visualization framework for interpreting reasoning in early-fusion VLMs.
- Designed ChartPADs, a persona-adaptive explanation framework that tailors chart descriptions to users' visualization literacy using an editor–critic multi-agent LLM pipeline (GPT-4o, Gemini 2.5 Pro).
- Conducted a multi-factorial mixed-design user study (50+ participants): personalized explanations improved comprehension accuracy by 22%, response confidence by 18%, and perceived clarity by 24%.
- Built an interactive web platform for explanation comparison and refinement loops.

Industry Experience

Data Analyst Intern, Microsoft Jan 2024 – Apr 2024
Cloud reliability analytics Hong Kong

- Designed classification pipelines for anomaly detection on Windows Defender and Azure error logs, improving early failure detection by over 15%.
- Conducted full-stack data analysis and automated reporting pipelines for service reliability metrics.

Software Developer, YSAIT Sep 2023 – Dec 2023
AI-driven e-learning platform Canada

- Built backend infrastructure for an AI-driven e-learning platform using Flask and the GPT-3.5 API.
- Developed an admin dashboard (Spring Boot, MyBatis, Redis, MySQL); reduced API response latency by 30% through caching and async pipelines; deployed on AWS EC2.

Awards & Honors

- **Undergraduate Student Research Award (USRA)**, University of Waterloo 2026
- **1st Place**, Scotiabank Data Science Discovery Days (team of 5, 500+ participants) 2023
- **Top 1%**, Euclid Math Contest (University of Waterloo CEMC) 2022
- **Gold Division, Full Marks**, USA Computing Olympiad (USACO), competed in C++ 2021
- President's Scholarship of Distinction, University of Waterloo 2022
- Faculty of Math Entrance Scholarship ($25,000 CAD), University of Waterloo 2022
- Statistics & Actuarial Science Competition Prize, University of Waterloo 2022

Service & Activities

- Student Volunteer, IEEE VIS 2025, Vienna, Austria Nov 2025
- Participant, Google–UWaterloo Futures Lab (Fall 2025 cohort) Oct – Dec 2025

Technical Skills

Languages: Python, C, C++, Java, R, SQL, JavaScript, TypeScript
AI / ML: PyTorch, TensorFlow, LangChain, vLLM, NumPy, Pandas, OpenCV
Frameworks: React/Preact with TypeScript, Vue, Spring Boot, FastAPI, Flask